

Mofe Omolade

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Computer Engineering

Vancouver, BC

Expected May 2029

EXPERIENCE

Electronics Engineering Intern

May 2026 – August 2026

Fulcrum Air

Calgary, AB

- Owned the complete design lifecycle of a 4-channel motor controller PCB to drive 12V actuators at 3A
- Designed system architecture around Teensy 4.1 supporting 150 W continuous power draw, multichannel motor control, CAN communication, analog current sensing, Hall-effect positioning, and operation on 240kV power lines
- Performed high-density routing for an 8-layer PCB within a strict 80 mm x 54 mm package, implementing orthogonal routing, dedicated power and ground planes, ground stitching, and 2.1 MHz switching node shielding
- Engineered a power-loss detection circuit and falling-edge ISR using a voltage divider and bulk capacitor, guaranteeing non-volatile EEPROM backup of actuator positions within 5 ms of power disconnection
- Built a test bench and developed C++ firmware to validate PCB function, verifying MCU pinouts, voltage regulation, inrush current stabilization, CAN signal integrity, and bidirectional actuation under dynamic loads

Electrical Team Member

September 2025 – August 2026

UBC Rover

Vancouver, BC

- Designed power architecture, telemetry systems, and motor drive PCBs for an autonomous test rover
- Tested an off-road powertrain prototype around the L298N full-bridge motor driver and Nucleo F446RE MCU
- Developed C firmware using STM32 HAL for UART communication, command parsing, and Hall effect feedback
- Performed stress testing to validate thermal regulation and electronic reliability in environments over 35°C

Electrical Engineering Research Assistant

July 2023 – August 2023

University of Calgary

Calgary, AB

- Modeled and simulated circuits in LTspice to support research on novel Op-Amp-based oscillator topologies
- Validated circuit performance, achieving 95% correlation between simulation and lab measurements
- Utilized oscilloscopes, signal generators, and DMMs to debug test circuits and verify frequency responses

PROJECTS

24V – 5V Buck Converter | LTspice, Altium

August 2026 – Present

- Designed an asynchronous buck converter powered by a 6S LiPo input to deliver up to 500 mA at 85% efficiency
- Implemented a high-side N-channel MOSFET with custom voltage-mode control loop, sizing propagation delays and slew rates to prevent shoot-through and minimize switching losses
- Simulating closed-loop compensation in LTspice, validating small-signal Bode stability and dynamic load response
- Performing 4-layer PCB layout in Altium Designer, tightly constricting high- di/dt switching loops, optimizing Schottky diode return paths, and routing unbroken ground reference planes to minimize EMI and radiated noise

Self-Balancing Robot | C++, ESP32, I2C

July 2026 – August 2026

- Developed a standalone C++ device driver for the MPU-6050 from scratch, writing directly to I2C registers to configure power states, set gyroscope and accelerometer ranges, and parse 14-byte data bursts
- Implemented non-blocking sensor data acquisition and applied a complementary filter to fuse accelerometer and gyroscope readings, achieving low-latency angle estimation while reducing gyroscope angular drift by 85%
- Engineering a custom closed-loop PID controller from first principles without third-party frameworks, incorporating anti-windup clamping, derivative filtering, and deadband compensation
- Programmed PWM generation using ESP32-C3 hardware timers to drive an L298N dual-motor bridge, converting raw PID outputs into directional motor commands

TECHNICAL SKILLS

Hardware: STM32, ESP32, CAN, SPI, I2C, UART, RS-485, PCB Design, FPGA Design

Tools: Quartus Prime, ModelSim, Git, MATLAB, Simulink, Altium, KiCad, LTspice

Languages: SystemVerilog, C/C++, Java

Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging